# Lecture 3

Tuesday, 29 August 2023 10:33 AM

## Infra-red spectroscopy

+ Identity of adsorbed species
+ Surface structures and phases
+ In-situ & operando techniques

Usually too stable to be an intermediate species

CO → 2143 $cm^{-1}$ (free)

### Frequency shifts & BE:

Stronger bond → Less energy to vibrate that bond → shift

↳ Antibonding

Backbond → need $e^-$

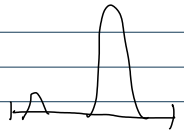
↑ wavenu. → stronger ↑ ???

$E = \frac{hc}{\lambda}$ $\frac{1}{\lambda}$ = wavenu.

$E = hc \cdot$ wave. no.

New MO formed when hy → M d orbitals interacting w/ $\pi$ orbitals in CO → It can happen that $\sigma$ bond could be de-stabilized more than the $\pi$ bonding

CO on Pt(111)




+ Area increase (due to coverage)
+ Shift due to interactions

### Operando transmission IR:

peak area correlates w/ time ; path length of gas should be minimized

Under reaction conditions

→ Add the tracer to know hydrodynamic vs. reaction kinetics

IR feature corresponding w/ something intermediate
↳ that is the intermediate

Intrinsic TOF → normalized TOF to something

to something

Steady state Isotopic Transient Kinetic Analysis (SSITKA)

The time constant for hydrodynamic (transport) constrictions is given by the carrier MS signal

# Lecture 4

Thursday, 31 August 2023 10:34 AM

Chapter 8 ; Niemantsverdriet ; Bulk vs. Surface-sensitive ;

## DRIFTS · Diffuse Reflectance IR (DRIFTS)

+ Transmission needs the IR to pass through ( self-supporting wafer required to be input in the sample)

+ But when support or catalyst is very dark & opaque → DRIFTS
  ie. powder properties
  ↓
  More towards the surface technique

+ Challenges : In-situ vs. operando
  ↓ reaction conditions
  ↳ Exactly how its done

+ Sample is "infinitely thick"
  ↳ But cannot be too thick since there will be significant gradients

Drifts issues : T gradients, diffusion resistance (Conc. gradient due to lack of porosity)

Drifts is not as quantitative as Transmission

+ Attenuated Total Reflectance (ATR) -IR:
  + Planar model catalysts, thin films
  + Water absorbs a lot of IR
  + Can be used for liquid-solids interfaces

+ UV-vis and Raman Spectroscopy

↑ wave → ↑ energy → ↓ wavelength

LMCT → Ligand-to-metal-charge-transfer

MLCT → metal-to-ligand "

d–d transitions : lower energy

CT → Higher

ε → Absorbsivities (Signal value)

Spectra → can you explain all the peaks & what peaks are not there.

## Raman Spectroscopy (RS)

## Raman Spectroscopy (RS)

Inelastic scattering; when light is incident

IR only gives single level transitions → Using FT to figure out which one got adsorbed

RS

+ choose a specific wavelength

- Rayleigh scattering
- Stokes shift
- Anti-stokes

Vibrational transitions not the same as electronic transitions

↓

Raman Shifts: you can figure out identity of material

## UV Raman

Removes flourescence of adsorbed Carbon species

# Lecture 5

Tuesday, 5 September 2023 10:32 AM

## Crystal lattice Structures (XRD, bulk characterization)

→ Next atom neighbors, → affects interactions w/ other atoms

→ Crystalline → regular order (zeolite)

Amorphous → No repeat unit; local disorder; glass, sand.

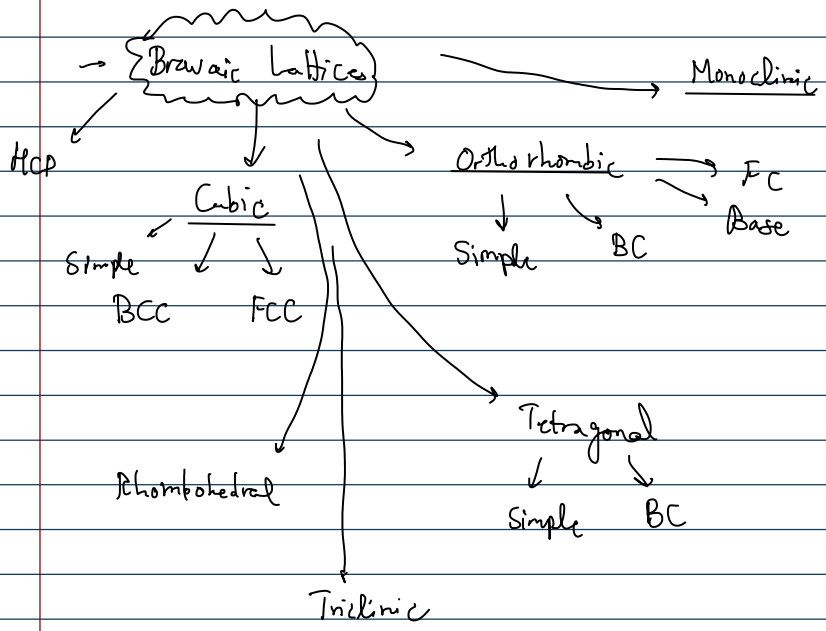
$SiO_2$ thin-film on Ru(0001) → first STM image of amorphous $SiO_2$

→ Study of active sites on amorphous silica is a challenge

→ lattice

→ 14 possible arrangements of points in space possible

→ Surface termination → a specific bulk arrangement will always end/ have specific surface termination




Axis are defined along the unit cell edges (and not cartesian)

[uvw] → no. of lattice steps taken to reach desired points

↳ These are vectors
so can be reduced to that

↓

But not that useful (we care about surfaces)

↓

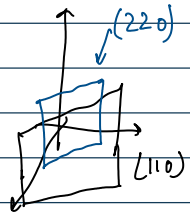
Crystal plane have <u>miller indices</u>

(hkl)

→ Intersection (has a value) of to axis

→ Parallel to an axis (has 0 value)

} Just is inverse Lie.




Hexagonal symmetry: Miller-Bravais (hkil)

$i = -(h+k)$

Maximize Coordination no. in bulk (FCC, HCP) ; Is Au, Pt FCC?

↓

?

→ Linear density

Planar density

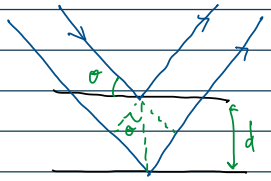
# Lecture 6

Thursday, 7 September 2023 10:35 AM

## X-ray Diffraction

+ Determine structure of material w/ some long-range order

+ Bragg's law: $\lambda = 2d \sin\theta$



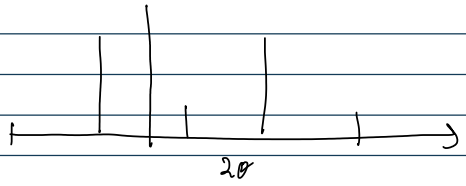

This is fixed → what we need to determine

Specific θ value for a given X-ray source

+ Geometry & intensity will vary on the source (synchrotron vs. lab based)
↓
Quantitative, under reaction conditions

+ Understanding XRD patterns:

$$d_{hkl} = \sqrt{\frac{a^2}{h^2+k^2+l^2}}$$




hkl (100) (110) (111) ....

Intensities of peaks decreases

For simple cubic → certain peaks show up

FCC → 111, 211, 311

BCC → 110, 220

} Based on the group you will have different lines that show up.

Extinction Rule:

Form factor: Scattering by an atom over scattering of an e⁻

↳ This is for each atom & heavier elements will have higher form factor

d spacing is related to the atomic size → so you do know which structure is diffracting in what pattern

We get zero information about the degree of crystallinity

Simulation of XRD patterns → can do that!

→ Rietveld Refinement → when you make a new structure, you use least squares fitting approach

→ Sherrer Equation → Instrument broadening is an issue

$$B(2\theta) = \frac{K\lambda}{L\cos\theta}$$

→ Laue Equations →

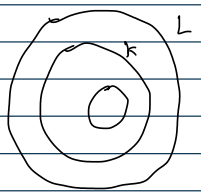
# Lecture 7

Tuesday, 12 September 2023 10:29 AM

## EXAFS & XANES } XAS

Structural info:
- + Oxidation state → mixture will be needed to sort out
- + Local geometry → XRD & planes; but this is photoelectrons (scattering electrons)
- + Surface coverage of Adsorbates
- + Valence orbital energies → This is where the chemistry happens

→ Build your own reactor to get in-situ data




K-shell (1s)

L (2s, 2p)

M (3s, 3p, 3d)

Energy of $e^-$ it feels changes w/ the nucleus

↓

photons excite the $e^-$ (1s, 2p $e^-$ most likely → core $e^-$) } photoelectrons

↳ How do you make sure?

↳ You guess it based on your guess of what the element is & its concentration } You want to figure out if the metal is bonded & to what (oxidation states, etc..)

+ Can use for amorphous or crystalline materials

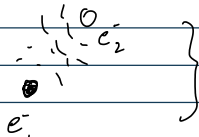
d-orbitals are doing the catalysis!

XAS:

Edge = peak

XANES → photo excitation (to unfilled orbital → via selection rules)

EXAFS → scattering (i.e. scattering due to greater than 1E $e^-$ interacting w/ neighbors)




} Interference pattern

λ, distance, scattering efficiency

XPS → KE of photoelectron → records the $e^-$ energy

XAS → records the photon energy (absorbed vs. not)

} Both give local structure!

Measure no. of photons in ion chamber (via charged plates)

$\log\left(\frac{I_0}{I_T}\right)$ ⇒ Absorbance

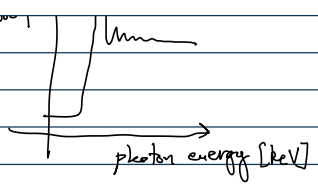
Flourescence (is just the inverse of absorbance)

Scattering is from neighbors!

Info:
- + No. of neighbors
- + Distance b/w atoms
- + Type of neighbor

Absorption

- Distance b/w atoms
- Type of neighbors
- Oxidation state




Diffraction needs long range order for interference patterns of x-rays

FT of Exafs →

# Lecture 8

Thursday, 14 September 2023 10:34 AM

**Propane Dehydrogenation** [Pd-Zn catalyst
↓
surface monolayer

C-C activation is more favorable (higher equilibrium conversion at lower temp.)

Structure sensitive vs. Structure insensitive
↓ (Dehydro)
reaction at vertex/edge

C-C bonds are broken at np. surfaces (not edges) → Structure sensitive

Zn, Ga, In, Sn, Cr → other alloys used w/ Pt

Pd is awful at PDH → Pd FCC

→ $\beta_1$-PdZn (Intermetallic) phase } has specific atomic distance (b/w atoms) → This has isolated Pd sites

↓

So thought that it would be good for catalysis } Looking at diff. alloys on the phase diagram chart

↓

Make core-shell structures a lot

EXAFS → coordination no.
Δ bond distance b/w atoms

Alloy formation as function of time
(when is the β phase formed)

**Electronic Effect:**

d-band center used as descriptor for reactivity

# Lecture 9

Tuesday, 19 September 2023 10:35 AM

## Temperature-programmed Techniques

TP

- R → Reduction
- O → Oxidation
- S → sulfidation
- D → Desorption
- SR → surface reaction
- RS → reaction spectroscopy

Intent behind the experiment

- track reduction/oxidation state of some part of solid → and then characterize the products given out
- specific use of $H_2S$ to reduce
- displacement or temperature control

## Info gained

+ Desorption vs. Adsorption energy : strength of binding (physisorption vs. chemisorption)
+ TPSR → rates of reaction & steps ; types/barriers of reaction
+ State of catalyst site (ion vs. metal state)
+ Quantify no. of sites (via TPD)

Multiple active sites ? → powdered, porous material complicates TPR

↳ Temp. peak does not always correspond to reaction rate of active site

TCD → thermal Conductivity detector (b/w reference & analyte stream)

[Wheatstone bridge]

↳ Gas composition affects the resistor resistance (due to change in thermal conductivity)
↳ changing current
↳ b/w reference & analyte

**MS** → sorting ions based on mass/charge ratio

Ionization

- electron impact → Large molecular fragments
- Chemical Ionization (soft ionization) → electron transfer through collision of molecules

Parention → fragments due to cleavage → major ion & then H

# Lecture 10.

Thursday, 21 September 2023 10:37 AM

## Temp. programmed Surface reactions

TPR → Temp. prog. reduction

Reduction mechanisms

- Shrinking core ($H_2$ fast dissociation)
- Nucleation & growth ($H_2$ dissociation slow) (stable oxides)

} formed an oxide & now need to reduce it from metal (+ water)

+ Linear ramp rate
+ $H_2$ consumption

} Experimentally?

$dT = \beta dt$

$r \propto \beta$

[TCD or MS] → to measure $H_2$ consumption rate

1) Assumption → model does not correspond w/ temp ramp → Is that true → does a particular oxide reduce in a particular way.

You can resolve peaks by using wet & dry reduction to shift equilibriums as recorded.

TGA - DSC

↑ tells you if the weight loss was exothermic or endothermic

TPSR →

Hydrocarbons serve as reductants for $NO_x$

i.e. provide H to break / reduce $NO_x$

# Lecture 11

Tuesday, 26 September 2023 10:36 AM

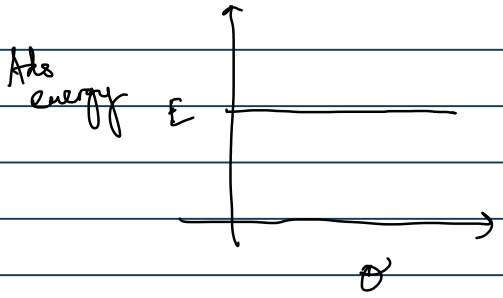
## Temperature - programmed Desorption

$NH_3$ → adsorbs on acids
↳ Is a base

LA → accepts $e^-$ from $NH_3$
(Lewis acid)

## Adsorption (binding to active site)

1. Langmuian : Ads. is independent of coverage to 1 monolayer (uniform)




+ No lateral interaction
+ "Single site" surface

Non-uniform surfaces
(strongest binding sites are populated first)

Adsorbate interactions → Temkin
→ Freundlich

2nd order + coverage dependent rate of desorption.

# Iglesia lecture

Tuesday, 3 October 2023 10:29 AM

$C \equiv O + O=O \rightarrow O=C=O$

+ Breaking into smaller steps (elementary steps) w/ smaller

+ Dimensions = vibrational modes of a molecule $(3N-1)$ } Becomes an N-dimensional space problem

## Mechanism-Based Chemical Kinetics

i) Determine kinetic response "reactive condition" $(P_j, T)$
↳ transport artefact
(probe away from actual active site so need to account for it)

$f(P_j, T)$

↳ Thermodynamics

$A \rightarrow B$ ← measure the rate of only forward reaction (isolate primary & secondary reactions)

2) Propose : elementary steps
↳ Identify reactive (& less reactive) species
↳ PSSH
(pseudo ss) } eliminate the need to know what you cannot see
↳ QE
↳ MASI (limited landing place) → empty, reactant, product

3) Assess reasonable : (seeking contradictions)
↳ Consistent w/ thermodynamics, calculations
(TST) → Thermo construct, theoretical in nature

↳ Direct observations (spectroscopic method)

$CH_4 + H_2O \rightarrow 3H_2 + CO$

Langmuir Hinshelwood reaction
vs.
Eyling ...

Elementary step } "Most simple"

vs.

Elementary step (when do you stop breaking it down) } "Most simple"

$aA + bB \rightleftharpoons cC + dD$

Eyling ...

↓

Has to occur as written! (can extract rate information directly) } Can use law of mass action

$$\frac{k_1}{k_{-1}} = K_1$$

$(*) + (A*) \rightleftharpoons (B*) + (C*)$

$\vec{r}_1 = k_1 (A*)(*) = k_1 (A*)\,\theta_*$

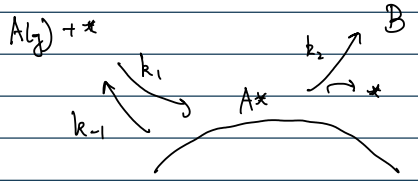
$(L)$ → total no. of squares

} Langmuirian treatment (catalyst is checkerboard)
+ Identical binding
+ Species are mobile → Langmuir (worked for GE)

↓

an approximation

$A \rightarrow B$




$\sigma_i$ → No. of times that step needs to take place

↑ stoichiometric constant

$A + * \underset{k_{-1}}{\overset{k_1}{\rightleftharpoons}} A*$

$A* \xrightarrow{k_2} B + *$

$(L) = (*) + (A*)$

$$\frac{d(A)}{dt} = -k_1 (A)(*) + k_{-1} (A*)$$

$$\frac{d(A*)}{dt} = k_1 (A)(*) - (k_{-1} + k_2)(A*)$$

using PSSH this term cannot change w/ time $\simeq 0$

# Iglesia lecture 2

Thursday, 5 October 2023 10:35 AM

NMR → tells you where in the molecule the isotope is.

Early on → used microwave rotational spectroscopy ( C-D rotation different from C-H)

Gringer : chemistry to make $^{13}C$ labels on carbon chains.

## Fisher-Tropsch synthesis

Chain initiation ; $CO + H_2 \rightarrow$ olefins + alcohols

↳ coming up w/ $C_1$ addition (one by one addition)

$R \xrightarrow{k_1} B^* \xrightarrow{k_2} P$ (Radioactivity was independent of chainsize)

$^{14}C$ is starting off & then carbons being added

rate constant ↑ intermediate ↑

SS rate: $r_{ss} = k_2 [B^*]$ } unable to distinguish

↳ transient experiments can decouple the k & $[B^*]$

↳ $\frac{d[P]}{dt} = k_2 [B^*]_{ss} \exp\{-k_2 t\}$

get this from decay

+ The "seed model" of Fisher-Tropsch

## Desorption / Adsorption Issue

↳ How do you stop molecules re-adsorbing?

Have isotopic gas flowing → displacing the $^{12}C$ on surface w/ $^{13}C$

## Oxidative Dehydrogenation of Propane

Homolytic C-H activation }

$$r = \frac{A}{[C \cdot D]}$$

TS of RDS step

C & D ↓ ↓ averages

RDS step is a step of point of no return

↓

(while the other elementary steps will be at equilibrium)

Lecture 14?

Tuesday, 17 October 2023 10:37 AM

## Solid state NMR (very different from liquid state)

ESR
↳ electron spin resonance
very sensitive but only applicable to certain applications

+ Low frequency & energy
+ Exciting spin states of nuclei (& $e^-$) } least amount of energy required but need sensitive detectors.
↓
splitting degenerate spin states
(need to actually spin the sample

### Larmor frequency

frequency of precession of $^1H$ ; $\omega = -\gamma B$

$\omega$ → frequency
$\gamma$ → gyromagnetic ratio $\propto \frac{1}{\text{mass of nucleus}}$

$$\gamma = \frac{-eg}{2m}$$

$I$ → nuclear spin quantum no.

NMR Receptivity

# Lecture 15

Tuesday, 24 October 2023 10:40 AM

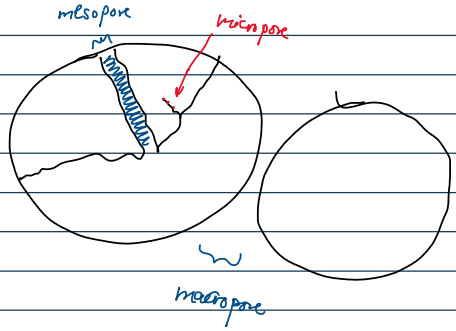
## Characterisation Methods for Surface area & pore size determination

Micropore, Mesopore, Macropore

Adsorption isotherm

Small angle X-ray scattering:

Mercury porosimetry: non-wetting liquid → pressure needed to fill the pore space




Gas adsorption: surface atoms → unbalanced forces / undercoordinated bonds

Physisorption forces: + Van der waal's forces → dispersion force
→ ion-dipole
→ ion-induced dipole
→ Dipole-dipole

→ non-specific (Can adsorb anywhere on the surface)!

→ How do you determine if it is blocking multiple sites??

→ Can also have multilayer (partial coverages)!

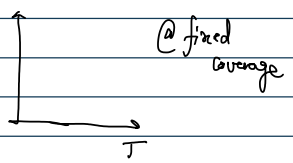
→ But may not be layer by layer } Need a mathematical model

All equilibrium measurements

→ isobar (p = const)

→ isotherm (T = constant) → coverage as function of increasing pressure

→ isostere (Coverage constant)

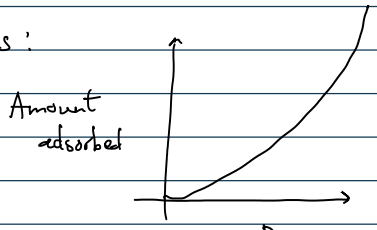



Use different adsorbing molecules and then coverage will be off by factor of molecule volume

# Lecture 15

Thursday, 26 October 2023 10:35 AM

Adsorption Isotherms:




Adsorbate-adsorbent interactions (favorable) ✓ } Presence of other adsorbates increases coverage

$N_2$ → quadrupole moment

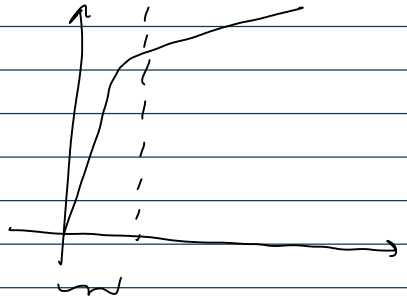
Hysteresis: material remembers memory of what the adsorption was.

MOFs: BET equation is not always valid for them

↓

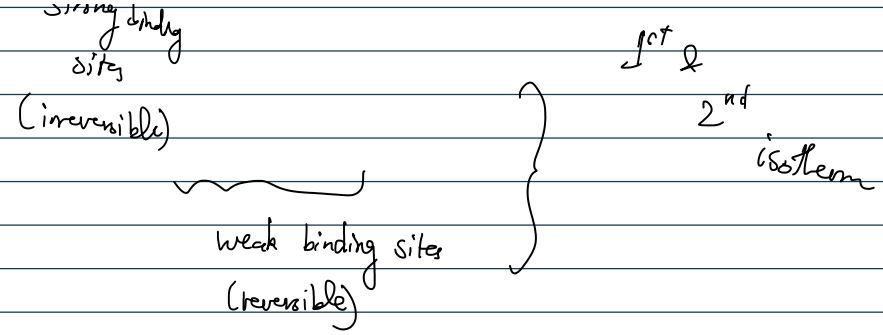
confusion b/w mesopores & micropores by the equation.

Chemisorption: quantifying surface atoms/ active sites (not just how many pores we have)



Strong binding sites

1st &




Complications: + Changes in surface due to adsorption

Support vs. Metal:

→ Ads on support vs. on weak site on support

→ Exp.: Henry's law from CO on support
(slope)
↳ Then compare slope w/ ads. slopes to say from support or metal.

Adsorbate-induced reconstruction → can draw the metal out!

Tuesday, 31 October 2023 10:37 AM

# Elemental Analysis

+ Atomic Absorption Spectroscopy (Bulk technique)

+ ICP

Analyzer: AES, OES, MS → Ar used fuel source

MS > AES, OES } MS has 3 order of magnitude resolution

Single wavelength → monochromator

Atomic Absorption Spectroscopy (AAS): vs. ICP → plasma ionizes & excites every e⁻

+ Solid samples are digested → & then atomized in chamber

↓

Lamp generates only specific wavelengths (instead of ICP)

## Kohn Sham Theory:

KE + Coulombic energy + Exchange correlation functional

(usually what we don't know & we approximate it)

Functional derivative w.r.t density

↓

density to minimize the expression

↓

Kohn-Sham orbitals (approx. to true one-electron orbitals)

" eigenvalues (approximations to the energies of the various one-electron orbitals)

## Nørskov d-band Theory (Newns-Anderson Theory)

+ Approximate wavefunction of AB molecule using tight-binding theory

A B

$\Psi_{AB} = c_A\Psi_A + c_B\Psi_B$

where, $\left(-\frac{1}{2}\nabla^2 + v_A\right)\Psi_A = \varepsilon_A\Psi_A$ → one-electron orbital for A

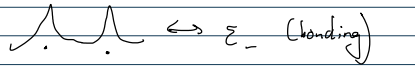
one-electron Hamiltonian: $H = -\frac{1}{2}\nabla^2 + v_A + v_B$

(neglecting $e^- - e^-$ interactions)

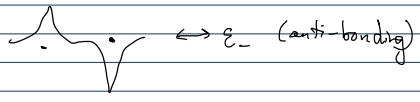
$H\Psi_{AB} = \varepsilon\Psi_{AB}$

↓ Math

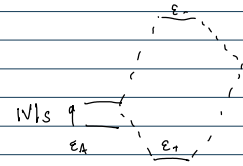
get two solutions:



$\leftrightarrow \varepsilon_+$ (bonding)



$\leftrightarrow \varepsilon_-$ (anti-bonding)




$S = \int \psi_A\psi_B$ } accounts for geom.

$\Delta E \approx 2\varepsilon_+ - \varepsilon_A - \varepsilon_B = -2SV - \sqrt{4V^2 + (\varepsilon_B - \varepsilon_A)^2}$

- $-2SV$: Pauli repulsion (positive)
- $\sqrt{4V^2}$: Covalent contribution to hybridization energy
- $(\varepsilon_B - \varepsilon_A)^2$: Ionic ant to hybridization energy

} Both ionic & covalent attractions matter

Newns-Anderson: Single adsorbate orbital interact w/ single orbital from $e^-$ in solid surface

$$n_a(\epsilon) = \frac{1}{\pi} \frac{\Delta(\epsilon)}{(\epsilon - \epsilon_a - \Lambda(\epsilon))^2 + \Delta(\epsilon)^2}$$ ← result

$\Delta(\epsilon)$ → density of states

- narrow → than adsorbate orbital will be split into bonding & antibonding
- broad → than adsorbate energy level is simply broadened (S & p) (Contribute a constant shift in energy)

Distribution as to why d-orbital has a narrow

↳ S & p has similar energies across all elements → but d bands very a lot and tend to form hybridised orbitals

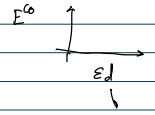
Can integrate DOS to obtain total energy of the system

↳ metal DOS are very narrow (d-bands)

$$\Delta E = -(1-f)\left[\sqrt{4V^2 + (\epsilon_d - \epsilon_f)^2} - (\epsilon_d - \epsilon_f)\right] - 2(1-f)SV$$

$f$ ← occupancy of fermi level

Non-linear curve but has some linearity




d-band center shift

Metal thin-films on substrates:

+Shifts in BE via shifts in d-band centers!

Hammer (1996)
Ruban (1997)

↳ Depositing what on what will

## Problem 1: TPD Analysis

Thermal desorption of CO from Rh(111) ; 10 K/s ; Coverage dependent desorption

Redhead equation: $E_{des} = RT_{max}\left[\ln\left(\frac{\nu T_{max}}{\beta}\right) - 3.46\right]$ ; $\beta = 10\ K/s$ ; $R = 0.0082\ kJ/(mol\ K)$

(a)

| $T_{max}$ / K | $10^{13}\ s^{-1}$ | $10^{14}\ s^{-1}$ | $10^{15}\ s^{-1}$ |
|---|---|---|---|
| 510 | 127.16 | 136.78 | 146.41 |
| 400 | 98.93 | 106.49 | 114.04 |
| 300 | 73.49 | 79.16 | 84.82 |

↳ Activation energies in kJ/mol

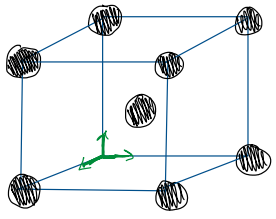
(b) They would be equal if we assume no interactions b/w adsorbate species ↗ reversibility b/w ads and des.

(c) The preexponential factor is chosen such that it is independent of coverage.
A choice of higher values indicate that there is a greater change in entropy b/w the molecule in ground state & transition state.

## Problem 2: Crystal Structures

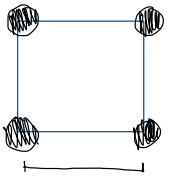
BCC ; Rhombic dodecahedral crystal shape

(i)



Body centered Cubic

→ 100 plane



$a = \frac{4R}{\sqrt{3}}$ ; R = radius of Fe atoms

↑ unit cell length

0.126 nm

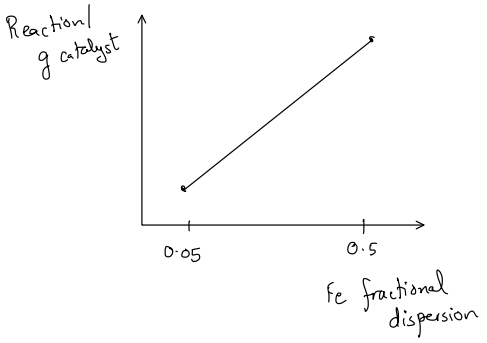
$a = \frac{4}{\sqrt{3}}(0.126 \times 10^{-7})\ cm =$

Exposed atoms per $cm^2$ : $\frac{1}{a^2} =$

For (110) Fe : $\frac{1}{\left(\frac{4R}{\sqrt{3}}\right)^2\ cm^2} = \frac{1}{\left(\frac{4 \cdot (0.126 \times 10^{-7})}{\sqrt{3}}\right)^2}\ \frac{atoms}{cm^2} =$ $\boxed{1.18 \times 10^{15}\ \frac{atoms}{cm^2}}$

(ii) CO hydrogenation ≠ f(Fe crystal size)

Fe dispersion varied from 0.05 → 0.50 } This results in more (exposed) surface Fe atoms to which CO can react w/




$$D_m = \frac{N_s}{N_T} = \frac{[\text{surface area}]\,(cm^2)\,(\rho_s)\,\frac{\text{atoms}}{cm^2}}{[\text{volume}]\,(cm^3)\,(\rho_v)\,\frac{\text{atoms}}{cm^3}} = \frac{8\sqrt{2}\,r^2}{\frac{16\sqrt{3}}{9}\,r^3}\cdot\frac{\frac{1}{r^2}}{\frac{1}{a^2}}$$

$$D_m = \frac{8\sqrt{2}}{2}\cdot\frac{9}{16\sqrt{3}\,r}\cdot\frac{a}{a}$$

edge length of dodecahedral

$$r = \frac{9\sqrt{2}}{4\sqrt{3}}\,\frac{a}{D_m}$$

$r = 1.07 \times 10^{-6}$ cm

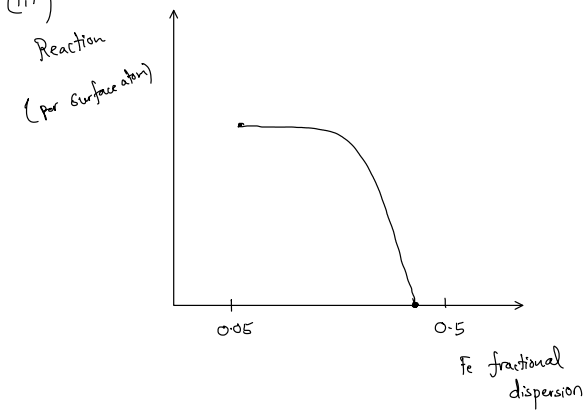
$= 1.07 \times 10^{-6} \times 10^{-2}$ m

$= 1.07 \times 10^{-8}$ m

$1.07\times10^{-7}\times10^{-2}$

| | | | | |
|---|---|---|---|---|
| | $r$ = 10.7 nm | @ | $D_m$ = 0.05 | |
| & | $r$ = 1.07 nm | @ | $D_m$ = 0.5 | → diameter range will be twice of this assuming $D = 2r$ |

(iii)

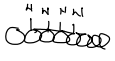



$NH_3$ reaction on surface atoms

↓

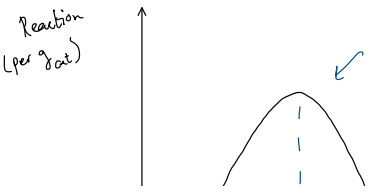
As dispersion ↑, surface atoms ↑

reaction rate accounts for increase in surface atoms

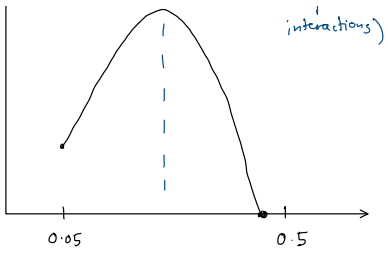



However, the rate decreases over time as the 4N sites are saturated → so not all dispersed sites will be used.

(especially as SA↓ so many sites are just blocked)




← Max rate where 4N adsorbate clusters are spread out (minimizing interactions)




(iv) $H_2$ chemisorption is not applicable on Fe since $H_2$ exhibits activated adsorption on Fe and thus will not give accurate results [Vanice]

Dispersion on Fe catalyst can be measured via CO chemisorption. CO adsorbs non-dissociatively but can adsorb linearly on single atom site or at a bridge where it binds to 2 sites

↳ Requires 2 isotherms for uptake for reversible uptake

CO chemisorption:

1) CO introduced and CO uptake ($\mu mol/g_{cat.}$) is measured as function of pressure
2) If support is present then blank runs w/ support are also measured
3) Reversible & irreversible uptake are deconvoluted
   - ↳ weak adsorption from support (Henry's law regime @ high pressures) ↳ Low coverage proportional to adsorbate pressure
   - strong chemisorption on reduced metal surface (Langmuirian isotherm → high pressure regime)
4) dispersion is calculated using total no. of atoms (metal loading) & the surface sites information from CO uptake
5) Average crystallite size is then calculated as $D = \left(\frac{0.6}{d}\right)\left(\frac{V_m}{A_m}\right)$ where for spherical particle assumption [Vanice]

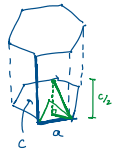
   D = dispersion, d = crystallite size, $V_m$ = bulk volume density, $A_m$ = bulk area density

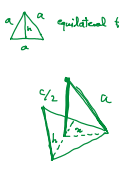
For TEM:

Given a histogram of # of particles w/ given size → the average can be calculated via a binomial distribution fit on the dataset.

Surface averaged & number-averaged crystallite size would be similar for materials where surface exposure is not reflective of its activity for chemisorption i.e. non-metallic sites or single atom sites

(v) Ideal HCP axis size ratio




equilateral triangle: $h = \sqrt{a^2 - \left(\frac{a}{2}\right)^2} = a\sqrt{\frac{3}{4}}$




$x = \frac{2}{3}h = \frac{1}{\sqrt{3}}a$

$\therefore \frac{c}{2} = \sqrt{a^2 - x^2}$

$= \sqrt{a^2 - \frac{1}{3}a^2}$

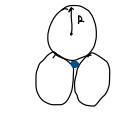
$\frac{c}{2} = a\sqrt{\frac{2}{3}}$

$$\frac{c}{a} = \sqrt{\frac{8}{3}}$$

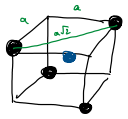
Interstitial sites exist b/w the spaces w/n a unit cell.

Closed Cubic Cell (CC) has { 4 octahedral sites (w/ coordinats to 6 atoms) & 8 tetrahedral sites (w/ " to 4 atoms)

HCP has { 6 octahedral sites 1/
12 tetrahedral sites

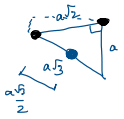



● → Tetrahedral site w/ radius r




$2R = a\sqrt{2}$

$R = \frac{a\sqrt{2}}{2}$




$(R+r) = \frac{a\sqrt{3}}{2}$

$\therefore \frac{a\sqrt{2}}{2} + r = \frac{a\sqrt{3}}{2}$

$r = \frac{a(\sqrt{3}-\sqrt{2})}{2}$

$r = 0.159\,a$

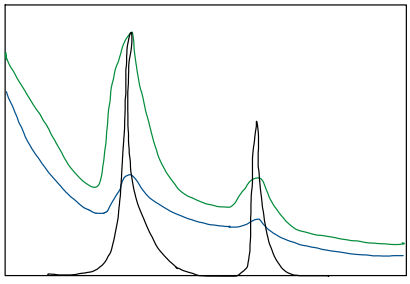
or $\boxed{r = 0.225\,R}$

For an atom to occupy a tetrahedral void, its radius must be 0.225 times the radius of Fe

HCP and FCC crystal structure are quite similar [(where the FCC(111) plane corresponds to the closed packed plane of HCP (001))] } Therefore the voids are similar

## Problem 3: XRD Analysis

(a)



— Pd reference
— 2.4 wt%. Pd loading on silica
— 1.1 wt%. Pd loading on silica

+ Reference signals are higher intensity & lower FWHM since Pd is a crystalline sample & will diffract x-rays effectively

+ $SiO_2$ is amorphous and therefore lower wt loadings will give less crystalline (more amorphous signals)

(b) (111) plane:

$n\lambda = 2d\sin\theta$ ; $2\theta = 40.2°$
$\lambda = 0.154$ nm for Cu Kα
$n$ = order of reflection ($n=1$)

$$d = \frac{(0.154\text{ nm})}{2\cdot\sin(40.2°/2)}$$

$d_{111} = 0.224\ nm$

(c) Pd lattice constant => 0.389 nm ; Pd is simple cubic

$$\left(\frac{1}{d_{hkl}}\right)^2 = \frac{h^2+k^2+l^2}{a^2}$$

$$\left(\frac{1}{d_{111}}\right)^2 = \frac{1^2+1^2+1^2}{a^2}$$

Solving for $a^2$ → $a^2 = 3\cdot(d_{111})^2$

$a = \sqrt{3}\cdot d_{111}$

$a = 0.388\ nm$ ✓ which is almost equal to 0.389 nm

(d) If using $M_o\ K\alpha$ → $\lambda = 0.07\ nm$

For $d_{111}$ : $n\lambda = 2d\sin\theta$

$$\frac{0.07}{2}\cdot\frac{1}{d_{111}} = \sin\theta$$

$\theta = 8.8°$

$2\theta = 17.98°$ for (111)

Similarly, $d_{200}$ :

$$2\theta = \sin^{-1}\left(\frac{0.07}{(2.)(0.194)}\right)\cdot 2$$

$2\theta = 20.79°$ for (200)

(e) $\langle L\rangle = \frac{K\lambda}{\beta\cos\theta}$ → Shemer equation

According to the Shemer equation → measuring at smaller wavelengths gives sharper peaks since $\lambda$ is smaller and diffraction lines shift to lower angles decreasing the $1/\cos\theta$ term (as shown in (d))

↳ Both effects reducing line broadening → and smaller particles can be probed } more resolved peaks

## Question 4

$H_2$ chemisorption on 1.5% $Rh/Al_2O_3$ catalyst

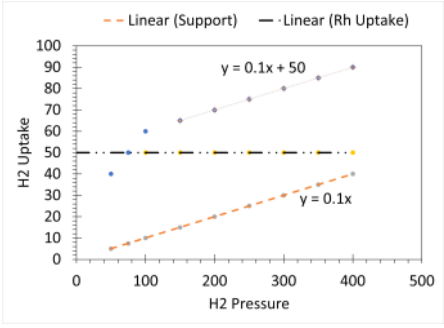
$$H_2 + 2Rh_s \rightarrow 2\ Rh_s - H$$

($Rh_s$: Rhuthenium surface atoms)

Assuming Rh does not form bulk hydride phase: we can separate the graph into 2 parts

1) Henry's Law regime → weak reversible adsorption on support
2) Adsorption on only Rh crystallites

| H2 Pressure (Torr) | H2 Uptake (umol/g_cat) | Support uptake | Difference ("a") |
|---|---|---|---|
| 50 | 40 | 5 | 35 |
| 75 | 50 | 7.5 | 42.5 |
| 100 | 60 | 10 | 50 |
| 150 | 65 | 15 | 50 |
| 200 | 70 | 20 | 50 |
| 250 | 75 | 25 | 50 |
| 300 | 80 | 30 | 50 |
| 350 | 85 | 35 | 50 |
| 400 | 90 | 40 | 50 |




y-intercept → gives the saturation adsorption on Rh crystals

$50\ \mu mol / g\text{-catalyst}$

slope → henry law slope w/ intercept of 0 } uptake by support only

Difference is $H_2$ uptake by Rh → $50\ \mu mol / g_{cat}$

Coverage:

$1\ mol\ H_2 \rightarrow 6.022 \times 10^{23}$ atoms $\rightarrow 2$ H atoms

$H_2$ uptake for $1\ g_{cat} \rightarrow 50 \times 10^{-6}\ mol \cdot \left(2 \times 6.022 \times 10^{23}\right) \frac{atoms}{mol} = 6.022 \times 10^{19}$ atoms

$$Rh_s \Rightarrow 6.022 \times 10^{19}\ atoms / g_{cat}$$

Dispersion = $\frac{N_s}{N}$ ($N_s$: surface atoms; $N$: total no. of atoms)

$= \frac{Rh_s}{Rh_T} = \frac{6.022 \times 10^{19}\ atoms}{(8.778 \times 10^{19}\ atoms)} =$

$D = 0.69$

$1\ g_{cat} \rightarrow$ has 1.5% Rh $\rightarrow$ 0.015 g of Rh

$\downarrow$

$MW(Rh) = 102.91\ g/mol$

$\downarrow$

$1.458 \times 10^{-4}$ mol of Rh

$\downarrow$

$Rh_T = 8.778 \times 10^{19}$ atoms

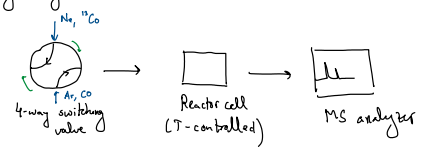
## Question 5

Method development for measure SA of Pt catalyst during WGS reaction via steady state isotopic transient kinetic analysis (SSITKA)

WGS reaction: $CO + H_2O \rightleftharpoons CO_2 + H_2$

Theory: Detecting time scale of evolution of labelled products & correlating w/ evolution of labelled surface active species (via time-resolved FTIR)

Gas delivery system:




→ We know CO adsorbs on Pt (usually linearly) → therefore we can calculate total moles of CO adsorbed on Pt by $^{12}CO/^{13}CO$ isotope switching to perform CO oxidation to get total surface area of catalyst

→ However not all surface sites are "active" for WGS reaction and therefore we can get idea about active metal sites via istopic transient measurements as such:

(1) Isotope tracer (Ne) & $^{13}CO$ labelled gas is introduced into reactor cell via the 4-way switching valve

(2) The mean surface residence time of $^{13}CO$ is calculated by integrating the area b/w Ne trace (which elutes first) and $^{13}CO_2$ (the WGS product)

(3) The surface conc. of Carbon-containing active intermediates is then calculated as the product of the rate & the mean surface residence time (which can be corrected for any support influences)

(4) This value can then be compared w/ the # of total surface atoms (via $H_2$ chemisorption). Assuming a certain no. of surface sites required to bind w/ the intermediate (i.e. one for CO binding w/ Pt via top site) an estimate of "active" surface area for WGS on catalyst can be determined.

## Question 6

Surface Area measurement via BET;

BET equation:

$$\theta = \frac{N}{N_m} = \frac{C\,(p/p_o)}{\left(1-\frac{p}{p_o}\right)\left(1-\frac{p}{p_o}+C\left(\frac{p}{p_o}\right)\right)}$$

; C = BET constant; $p_o$ = vapor pressure of liq. $N_2$; p = equilibrium pressure; N = Adsorbed gas [mol/g]; $N_m$ = Monolayer of adsorbed gas [mol/g] (theoretical)

↓ Linearized

$$\frac{(p/p_o)}{v\left[1-\frac{p}{p_o}\right]} = \frac{(C-1)}{v_m C}\left(\frac{p}{p_o}\right) + \frac{1}{v_m C}$$

; v = adsorbed gas quantity (mass)

BET plot: $\frac{1}{v}\left[\frac{p/p_o}{1-p/p_o}\right]$ vs. $\frac{p}{p_o}$ (for $0.05 < \frac{p}{p_o} < 0.35$)

Cell Volume:

$PV = nRT$; No. of moles in calibrated volume; $n = \frac{PV}{RT} = \frac{P_i \cdot (1801\ cm^3)}{RT}$

Assuming He does not adsorbed to sample → no. of mols of He should be equal to mol in calibrated volume + mol in cell

↳ Moles in cell build up since it is sequential pumping

↳ Assuming there is no heat transfer b/w cell reservoir & calibrate volume manifold we can get estimate of cell volume

↳ Assuming there is no heat transfer b/w cell reservoir & Calibrate volume manifold we can get estimate of cell volume

$N_2$ adsorbed :

Amount present @ initial + Amount introduced in cell volume = Amount adsorbed + Amount remaining (final)

Amount adsorbed = (Amount present @ initial + Amount introduced in cell volume) − Amount remaining

w/ sequential pumping: initial of $N_2$ are the no. of moles in manifold.

Amount adsorbed by sample = Total in Cell − remaining in cell

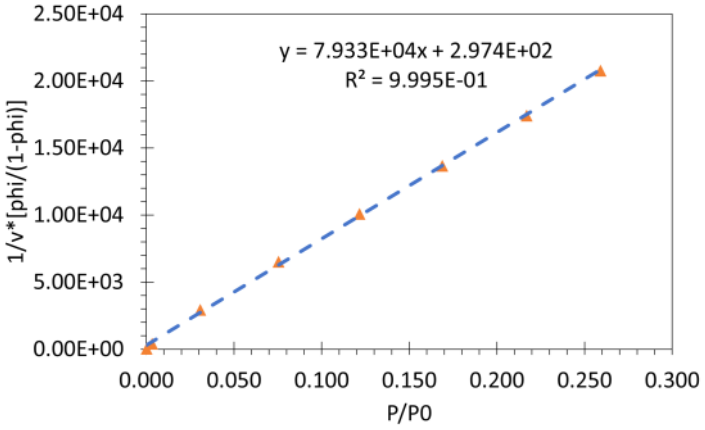
($N_2$ introduced (by pressure increase) in Calibrated volume + $N_2$ remaining in the cell from previous run + $N_2$ remaining in the calibrated volume after previous run − $N_2$ remaining after end of current run in calib. volume)

| Constants | | |
|---|---|---|
| R | 62.364 | L Torr/(mol K) |
| | 62364 | cm3 Torr /(mol K) |
| T | 21.2 | C |
| | 294.35 | K |
| T_cell | 77 | K |
| Calib. Volume | 18.01 | cm3 |
| P_0 (N2 @77K) | 101.3 | kPa |
| | 759.8 | Torr |
| Ax (ads. cross section) | 0.162 | nm2 |
| | 1.62E-19 | m2 |
| MW N2 | 28.0 | g/mol |
| N_Av | 6.02E+23 | atoms/mol |
| Molar Volume (N2) | 2.24E+01 | dm3/mol |
| | 2.24E-02 | m3/mol |
| Mass of sample | 2.60E-01 | g |

| Raw Data | | Calculated Values | | | |
|---|---|---|---|---|---|
| **P_i (Torr)** | **P_f (Torr)** | **n_He_i (mol)** | **n_He_f (mol)** | **n_cell (mol)** | **Cell volume (cm3)** |
| 19.6 | 4.8 | 1.92E-05 | 4.71E-06 | 1.45E-05 | 14.53 |
| 51.8 | 16.5 | 5.08E-05 | 1.62E-05 | 4.92E-05 | 14.31 |
| 68.4 | 29.4 | 6.71E-05 | 2.88E-05 | 8.74E-05 | 14.28 |
| 79.4 | 41.9 | 7.79E-05 | 4.11E-05 | 1.24E-04 | 14.24 |
| | | | | | |
| | | | Average | 14.34 | cm3 |
| | | | | | |
| | | | **Total Volume** | **32.35** | **cm3** |
| | | | (Calib. Volume + Cell volume) | | |

| Raw Data | | Calculated Values | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| P_i (Torr) | P_f (Torr) | N2 @initial (mol) | N2 @final_calib. vol (mol) | N2 introduced (mol) | N2 @final_cell volume (mol) | Initial (additive) | N2_total @cell (mol) | N2 adsorbed (cell) (mol) | N2 adsorbed cum. | Phi = P/P0 | 1/v (m3 @STP) | (1/v)*[(P/P0)/(1-(P/P0)] | *Check: (1/v)*[P/(P0-P)]* |
| 162.3 | 0.1 | 0.0001592334 | 0.0000000981 | 0.0001592334 | 0.0000002985 | 0.0001592334 | 0.0001591353 | 0.0001588368 | 1.59E-04 | 0.000 | 2.81E+05 | 3.70E+01 | *3.70E+01* |
| 207.5 | 2.5 | 0.0002035794 | 0.0000024528 | 0.0002034813 | 0.0000074636 | 0.0002035794 | 0.0002014252 | 0.0001939615 | 3.53E-04 | 0.003 | 1.27E+05 | 4.18E+02 | *4.18E+02* |
| 220.2 | 23.4 | 0.0002160394 | 0.0000229579 | 0.0002135867 | 0.0000698596 | 0.0002160394 | 0.0002005452 | 0.0001306856 | 4.83E-04 | 0.031 | 9.23E+04 | 2.93E+03 | *2.93E+03* |
| 235.9 | 57.3 | 0.0002314428 | 0.0000562173 | 0.0002084849 | 0.0001710664 | 0.0002314428 | 0.0002450850 | 0.0000740186 | 5.58E-04 | 0.075 | 8.01E+04 | 6.53E+03 | *6.53E+03* |
| 254.9 | 92.4 | 0.0002500838 | 0.0000906541 | 0.0001938664 | 0.0002758558 | 0.0002500838 | 0.0003304961 | 0.0000546403 | 6.12E-04 | 0.122 | 7.29E+04 | 1.01E+04 | *1.01E+04* |
| 289.4 | 128.3 | 0.0002839319 | 0.0001258758 | 0.0001932778 | 0.0003830335 | 0.0002839319 | 0.0004339119 | 0.0000508784 | 6.63E-04 | 0.169 | 6.73E+04 | 1.37E+04 | *1.37E+04* |
| 324.3 | 164.9 | 0.0003181725 | 0.0001617843 | 0.0001922967 | 0.0004923011 | 0.0003181725 | 0.0005394217 | 0.0000471207 | 7.10E-04 | 0.217 | 6.29E+04 | 1.74E+04 | *1.74E+04* |
| 336.6 | 196.9 | 0.0003302401 | 0.0001931797 | 0.0001684558 | 0.0005878356 | 0.0003302401 | 0.0006293615 | 0.0000415260 | 7.52E-04 | 0.259 | 5.94E+04 | 2.08E+04 | *2.08E+04* |

| BET Graph calcs | | |
|---|---|---|
| | | |
| slope | 7.93E+04 | |
| Intercept | 2.97E+02 | |
| v_m | 1.26E-05 | m3 |
| | | |
| Surface Area | 5.47E+01 | m2 |
| | | |
| **Specific SA** | **210.43** | **m2/g** |




From the BET plot:

Slope = $s$ ; intercept = $I$

$v_m = \frac{1}{s+I}$ ; $C = 1 + \frac{s}{I}$

$S_{total}$ = Total surface area = $\frac{v_m N_A A_x}{V}$

$A_x$: adsorption cross section of adsorbate (common value of $0.162\ nm^2$ is used for $N_2$)

$V$: molar volume of adsorbate gas; for $N_2$: $22.393\ \frac{dm^3}{mol} = 22.393 \times 10^{-3}\ \frac{m^3}{mol}$

$m^3 \cdot \frac{atom}{mol} \cdot \frac{mol}{m^3} \cdot \frac{m^2}{atom} [=] m^2$

Specific SA = 210.43 $m^2/g$

# Microscopy

Tuesday, 14 November 2023 10:36 AM

→ Measure of electron dose → on surface structure effects

→ Particles have different amount of facets → affecting reaction rate

EELS → electron energy loss spectroscopy

↓

zero-loss

Why electron microscopy?

e⁻

+ decrease in wavelength (have lower wavelength) → They are still electromagnetic radiation?

Birk → 0.65 Å resolution

+ e⁻ can exhibit particle-wave duality { diffraction, interference, photoelectric effect }

↓ wave ↓ particle

& e⁻ are scattered, photons are not → visible light spectroscopy

w/ gases